Sai Shanmukh Habbar Kalle

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Education

- **University of Petroleum and Energy Studies** — *Expected Graduation: May 2028*
  B.Tech, Computer Science and Engineering (**8.24/10**) — Dehradun, Uttarakhand

Experience

- **National Institute of Technology Karnataka, Surathkal** — *June 2026 – July 2026*
  Research Intern — Surathkal, Karnataka
  - Proposed a parameter-efficient speaker profiling framework by integrating Mixture of Recursions (MoR) with Wav2Vec 2.0 for joint age, height, and gender prediction.
  - Designed an adaptive recursive routing mechanism with shared Transformer blocks, reducing trainable parameters by **18.6%** while maintaining competitive performance on the TIMIT and NISP benchmarks.
  - Developed a gender-aware multi-task prediction framework using FiLM conditioning and uncertainty-weighted optimization, achieving **100%** gender classification accuracy on the TIMIT dataset.
  - Conducted multilingual and cross-dataset evaluation, authored a research paper, and released the complete implementation as an open-source PyTorch project.

Projects

Modular ML/DL Framework — Independent Research — *2025 – 2026*

GitHub — Python, NumPy

- Built a modular deep learning framework, replacing hardcoded network structures with a custom Autograd engine that uses a Directed Acyclic Graph (DAG) and topological sorting to compute gradients for arbitrary depths.
- Fixed multi-threaded state leaks and reduced inference memory footprint by 30% by refactoring activation caching. Added vectorized Adam and AdaMax optimizers with $1e^{-8}$ epsilon clipping to handle numerical stability.
- Added Batch Normalization and Dropout layers, implementing explicit state management to handle computational graph masking during training and moving average variance buffers during inference.
- Verified analytical gradients within a $1e^{-5}$ tolerance bound by implementing a automated testing suites to mathematically check custom derivations against finite difference approximations.

GameBoy Emulator — Independent Project — *April, 2026 – May, 2026*

GitHub — Java, JUnit

- Architected a functional GameBoy emulator from scratch in Java, translating raw Z80 hardware specifications and opcode tables into a complete CPU execution engine and memory management unit.
- Engineered the LCD controller (PPU) state machine to decode tile memory and map backgrounds, synchronizing JVM execution loops with the native 4.19 MHz CPU clock cycles.
- Verified core architectural behavior by writing a JUnit test suite to validate memory-mapped I/O, register state transitions, and instruction accuracy against known hardware edge cases.

Skills and Tools

- **Languages:** Java, Python, C, C++
- **Frameworks & Libraries:** PyTorch, NumPy, Pandas
- **Developer Tools:** Git, GitHub, Linux
- **Core Concepts:** Data Structures & Algorithms, Machine Learning, Concurrency, Systems Architecture